Q-Reflector: Learning GUI Action Reflection with Generated Conditions

Linqiang Guo, Wei Liu, Li Gu, Yang Wang, Tse-Hsun (Peter) Chen

Paper version: arXiv 2608.05587v2 · CUA Radar Reel poster, not made or endorsed by the authors

Problem

> *Frontier VLMs and specialized GUI models exhibit imbalanced recognition of valid and invalid transitions.*

Mobile GUI agents need a **post-action reflector** that judges each step.

Motivation: three open gaps

1. **Imbalance.** GPT-5.2: 95.10 negative vs 51.39 positive accuracy.
2. **Valid is not done.** A useful preparatory step must be accepted.
3. **Cost.** A reflection call after every action adds tokens.

Evaluation setup

AndroidWorld · offline | MobileWorld · online | Qwen3-VL-8B reflector | MAI-UI-8B · GUI-Owl-8B | 1,082 transitions | 117 tasks | 50-step cap

- **Offline** transition accuracy and **online** task success are kept apart.
- **Held out:** both test sets differ from the training benchmarks.

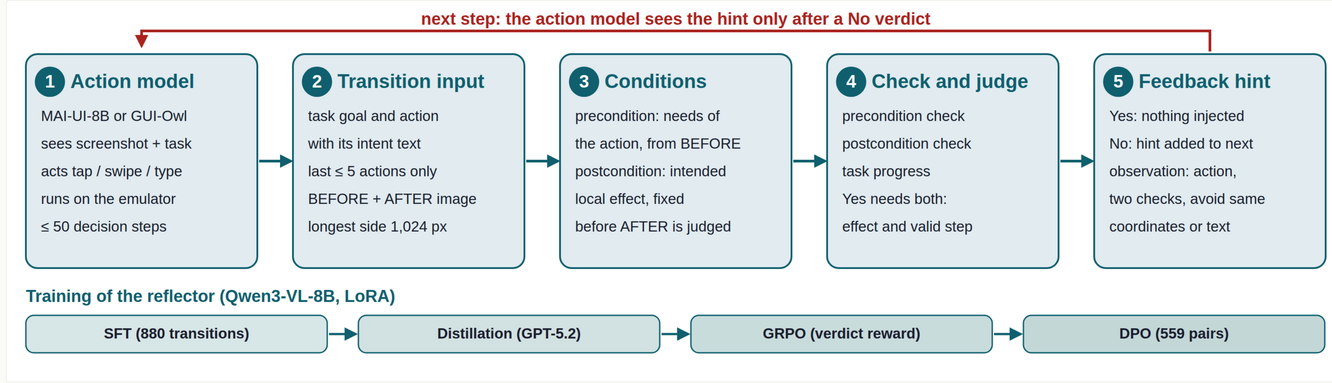
Our read (CUA Radar)

81	★★★	transfer
score of 100	band	evidence level

- **Reports:** online eval; separate test benchmarks; strong baselines; cost; ablation; judge check.
- **Not reported:** variance across runs; budget only partly matched.
- Pattern C18. Authors' numbers, not rerun.

Method: a trained reflector that writes conditions, then judges

- **Sees / acts:** task, action, ≤ 5 past actions, BEFORE + AFTER screenshots in; JSON verdict out.
- **Signal:** audited verdict labels; SFT, distillation, GRPO, then DPO.
- **Platform:** Android; separate module, action model unchanged.



Our redraw (not the paper's figure): the action loop with the post-action reflector and its three-stage training.

$$y_t = 1 \iff E_t \wedge V_t$$

E_t: intended local effect realized; V_t: valid step toward the task

Key results: MobileWorld online task success (%)

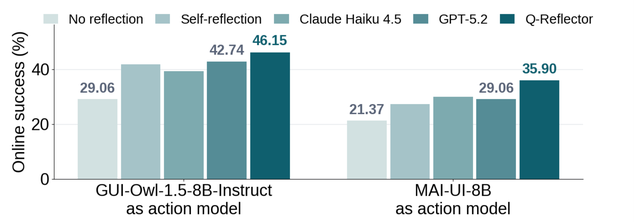


Table 2, redrawn: with condition generation, 117 tasks; GUI-Reflection-8B omitted.

+17.09 / +14.53 pts vs no reflection

SO WHAT → *Offline AndroidWorld accuracy is in the controls table.*

Ablation: condition generation, MobileWorld success (%)

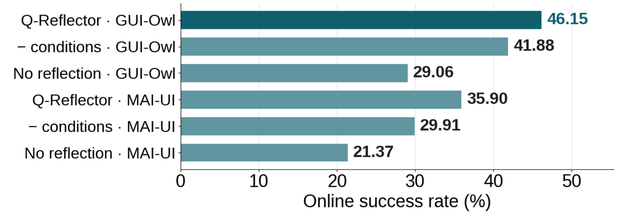


Table 2, redrawn: separately trained variants, same data and stages.

SO WHAT → *Conditions add 4.27 pts with GUI-Owl and seven tasks with MAI-UI. Input tokens rise 41.1% / 76.0%; output tokens fall 5.6% / rise 50.8%. The paper notes the benefit depends on the action model and reflector.*

Headline Numbers

46.15%
MobileWorld task success
vs 29.06% with no reflection

35.90% MAI-UI, MobileWorld
82.16% offline, AndroidWorld
+17.09 vs no reflection

The paper's own controls

CONTROL	Q-REFLECTOR	COMPARED WITH
OFFLINE accuracy	82.16 / bal. 81.45	GUI-Owl 77.36, GPT 70.33
OFFLINE pos. / neg.	86.79 / 76.12	GPT-5.2 51.39 / 95.10
Online, no cond. (MAI-UI)	29.91	Haiku 31.62, self 30.77
Online, no cond. (GUI-Owl)	41.88	GPT-5.2 40.17
Input tok./task (Owl)	236,132	no cond. 167,299

Takeaway

A trained 8B reflector lifts live MobileWorld success for two 8B action models; mobile only, 117 tasks, one run each.

Qwen3-VL-8B + LoRA | 49.91 GPU-h, H200 | 3.42 pts vs GPT-5.2

Scan to Read

Paper | Code